

BIOL-8 Comprehensive Final Exam — Answer Key

Modules 01–17 (cumulative) · 100 points total (51 + 15 + 25 + 9)

Part A: Multiple Choice (51 points)

Q	Ans	Brief rationale
1	C	Hypotheses drive testable predictions.
2	C	Cells → tissues → organs → organ systems.
3	D	Selection links survival/reproduction to inherited variation and allele frequencies.
4	C	Mass number = protons + neutrons (that isotope).
5	A	NaCl lattice = ionic attractions.
6	D	pH 11 is basic relative to pH 7.
7	D	Digestive cleavage with water = hydrolysis.
8	D	Steroids are lipids.
9	C	Primary structure = amino-acid sequence.
10	B	Rough ER = ribosomes on cytosolic face.
11	D	Lysosomes digest/recycle with acidic hydrolases.
12	B	Animal cells lack photosynthetic chloroplasts compared with plant mesophyll.
13	C	Fluid mosaic = bilayer + mobile proteins + surface diversity.
14	D	Osmosis = water across selective barriers.
15	B	Isotonic → little net water movement vs extreme hypo/hypertonic shifts.
16	B	ATP couples exergonic phosphate chemistry to cellular work.
17	C	Mitochondria host ETC + chemiosmotic ATP synthase.
18	B	Competitive inhibitors occupy the active site.

Q	Ans	Brief rationale
19	C	Translation = mRNA → polypeptide at ribosomes.
20	B	RNA polymerase runs transcription.
21	A	RNA uses uracil paired with adenine in standard contrasts with DNA/thymine.
22	B	Mitosis + cytokinesis → two diploid daughters (barring mutation).
23	A	Crossing over in prophase I of meiosis.
24	B	Meiosis halves chromosome number for gametes; fertilization restores diploidy.
25	B	Negative feedback reduces error around a set point.
26	A	Chewing = mechanical; amylase = chemical digestion.
27	D	Arteries carry blood away from the heart (veins return).
28	B	Alleles = alternative gene versions.
29	C	Complete dominance $Aa \times Aa \rightarrow \sim 3$ dominant : 1 recessive phenotypes.
30	C	AB blood shows both A and B antigens (codominant expression).
31	C	Adult hematopoiesis mainly in red marrow.
32	B	Skull, vertebrae, ribs = axial skeleton.
33	C	Freely movable lubricated joints = synovial (diarthroses).
34	C	Thick filaments = myosin.
35	C	Sliding filament: thin filaments slide, Z-discs move closer.
36	D	NMJ uses acetylcholine.
37	C	CNS = brain + spinal cord.

Q	Ans	Brief rationale
38	C	Axon carries signals away from soma.
39	D	Sympathetic mobilization → faster heart rate, blood shift toward skeletal muscle (overview).
40	D	Prokaryotes lack membrane-bound nucleus (nucleoid).
41	B	Bacteria typically divide by binary fission.
42	D	Viruses need host machinery to replicate.
43	A	Systemic venous return → right atrium first.
44	B	CO ₂ diffuses from pulmonary capillary blood into alveolar air.
45	C	Hemoglobin carries most O ₂ in RBCs.
46	B	Logistic growth slows near carrying capacity K.
47	B	Cleaner + client benefiting = mutualism.
48	D	Much energy lost as metabolic heat between trophic transfers.
49	D	Evolution = allele-frequency change across generations (populations).
50	C	Streamlined dolphin vs shark = convergent / analogous morphology.
51	B	Homology signals shared ancestry even if functions differ.

Part B: Fill in the Blank (15 points)

Blank	Answer	Notes
1	homeostasis	Stable internal conditions via feedback.
2	transcription	DNA → RNA synthesis.
3	translation	mRNA → polypeptide on ribosomes.
4	glycolysis	Cytosolic start of glucose catabolism.
5	meiosis	Halves chromosome number for gametes.
6	heterozygous	Two different alleles at a locus (Aa).
7	axial	Core skeleton: skull + vertebral column + rib cage framing.
8	osteoclast	Bone-resorbing cell.
9	synovial	Freely movable lubricated joints.
10	myosin	Thick filament motor protein.
11	acetylcholine	NMJ transmitter.
12	dendrite	Incoming signal processes.
13	peptidoglycan	Thick Gram-positive walls retain crystal violet introductory contrast.
14	capsid	Protein coat surrounding viral genome.
15	hemoglobin	O ₂ -binding pigment in erythrocytes.

Unused word-bank entries (seven decoys typical): alveoli, centriole, homology, logistic, omnivore, predation, sympatric—not required for blanks 1–15 above.

Part C: Integrated Short Answer Rubric (25 points)

Students submit five answers chosen from seven prompts ($5 \times 5 = 25$ max).

Grading. Score only **five** responses. Use each prompt's numbered rubric below. If question numbers conflict between the headline list and the **Question number** line on each **Response**, follow the labeled Response spaces (or reconcile with the student). Award **0** for a sixth graded response—extra answers beyond five are not scored.

Generic per-prompt ladder

Score	Descriptor
5	Accurate, specific vocabulary, logically ordered, satisfies every bolded element in the prompt.
3–4	Mostly correct with a minor gap OR thin examples but core idea intact.
1–2	Fragmentary understanding OR major misconception in one pillar of the prompt.
0	Off-topic, blank, or contradictory to basic course facts.

1. Mitosis vs meiosis (key points)

- Mitosis: one division cycle → two **2n** daughters ordinarily genetically matched (somatic renewal/repair/growth).
- Meiosis: two divisions → **four n** gametes with recombination/variation potential.
- Crossing over (**prophase I**) and independent assortment underpin meiotic genetic diversity contrasts.
- Cite plausible organ-level stakes (examples: epidermal turnover vs gamete formation for reproduction).

2. Central dogma + mutation (key points)

- DNA replication (optional brief), **transcription**, **translation** sequencing.

- Named machines beyond ribosomes plausible: helicase/topoisomerase, **RNA polymerase**, spliceosome RNA-processing if discussed, aminoacyl-tRNA synthetases (**if emphasized**), transcription factors regulating initiation, etc.—accept any **two legitimate distinct** players along the pathway.
- N nonsense/stop mutation can truncate or destabilize polypeptide → loss or gain of dysfunctional protein activity.

3. Sliding filament + denervation reasoning (key points)

- Calcium exposes myosin-binding sites on regulated thin filaments; **ATP** powers myosin cross-bridge cycling; filaments slide, sarcomeres shorten **Z-disc to Z-disc**.
- Motor-neuron lesion blocks **excitation–contraction coupling** at NMJ/end-plate—even with circulating ATP, skeletal fibers may not depolarize or recruit calcium properly to contract.

4. Pulmonary vs systemic + gas exchange (key points)

- Pulmonary loop: right heart → pulmonary arteries → lung capillaries → pulmonary veins → left atrium (intro map).
- Systemic loop: left ventricle → aorta → systemic arteries → capillary beds → systemic veins → right atrium.
- **Oxygen** loads from alveolar air into pulmonary capillary blood; **carbon dioxide** unloads from pulmonary capillary blood into alveolar air for exhalation.

5. Antibiotic resistance + selection (key points)

- Pre-existing **genetic variation** in tolerance.
- Drug imposes **selection**—tolerant survivors reproduce preferentially.
- Heritable resistance **spreads** clonally / via plasmids (if introduced) within hospital transmission networks—population trait frequencies shift, not training of individuals.

6. Carrying capacity and population growth curves (key points)

- **K** framed as approximate ceiling where births $\approx$ deaths / growth slows because limiting resources, space, waste buildup, etc. constrain increase (acceptable paraphrases).

- **Exponential:** growth rate high when far below limits; unchecked multiplicative phase in simple models (**J-shaped** acceptable wording).
- **Logistic:** growth slows markedly as $N \rightarrow K$; **S-shaped** saturation language; contrast “steep climb when small N ” versus “flattening near K .”

7. Population evolution and natural selection (key points)

- Evolution as **allele-frequency change in populations over generations**, not individual “improvement” or lifetime training alone.
 - **Natural selection:** heritable variants differ in survival/reproduction; advantageous traits tend to spread over generations (**frequency change** articulated).
 - Accept brief link to antibiotic-resistance vignette **only if** framed as populations, not individual learning.
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Part D: Essay (9 points)

Scoring guide

Score	Descriptor
9	Clear thesis, accurate multi-unit vocabulary, ≥ 2 specific examples, synthesis paragraph.
6–8	Solid structure; one weak example or missing one requested dimension.
3–5	List-like or thin integration; misconceptions limited to non-central details.
1–2	Mostly misconceptions or missing course mechanisms.
0	Blank or unrelated.

Option prompts — grading anchors

A — Ecology. Must address (J) vs (S) / logistic **K**, one density-dependent and one density-independent limiter, ecological roles (**producer/consumer/decomposer**) and why pyramids widen at basal production / energy dissipation heuristic.

B — Evolution. Must separate homology vs analogy with distinct examples; \n explain convergent evolution; \n tie isolation concepts to divergence/speciation vignettes (**intro** \n level).

C — Homeostasis. Must combine **negative feedback** with coordinated **nervous + endocrine** signals stabilizing glucose **or** temperature; \n name ≥ 2 **organs/tissues** with directional signaling logic.

End answer key